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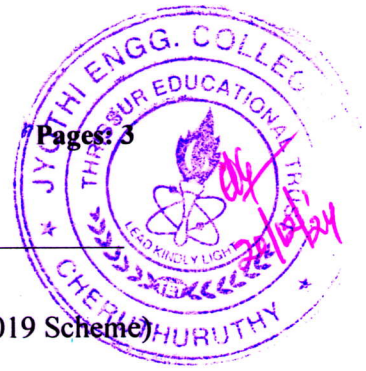
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APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

B.Tech Degree S6 (S, FE) / S4 (PT) (S, FE) Examination December 2024 (2019 Scheme)



Course Code: CST304

Course Name: COMPUTER GRAPHICS AND IMAGE PROCESSING

Max. Marks: 100

Duration: 3 Hours

PART A

Answer all questions, each carries 3 marks.

Marks

- 1 What is the purpose of frame buffer in a display system? (3)
- 2 The spatial resolution of an image is given by 1024 X 1024. What is the storage requirement if it is represented by 128 gray levels? (3)
- 3 Show that the composition of two successive rotations is additive by concatenating the matrix representations for $R(\theta_1)$ and $R(\theta_2)$. (3)
- 4 Magnify the triangle ABC with A (3, 3), B (8,2), C (3,9) to twice its size while keeping (3,3) fixed. (3)
- 5 Distinguish between object space and image space methods for visible surface detection. (3)
- 6 Is there any point at which a set of projected parallel lines appears to converge? Justify your answer. (3)
- 7 Define sampling and quantisation of an image. (3)
- 8 Explain the terms adjacency and connectivity in the context of digital images. (3)
- 9 Suppose an image is having a foggy appearance. What should be done with the gamma values to get a clearer picture? (3)
- 10 Write any two applications of image negatives. (3)

PART B

Answer one full question from each module, each carries 14 marks.

Module I

- 11 a) Explain the working of CRT with a neat diagram. (7)
- b) Perform scan conversion of the line segment with endpoints (30, 20) and (20, 13) using DDA line drawing algorithm. (7)

OR

- 12 a) Find the intermediate pixel positions for the line having end points (2,2) and (9,6) (7)
using Bresenham's line drawing algorithm.
- b) With a suitable figure, explain the shadow masking technique in CRT. (7)

Module II

- 13 a) Perform the following transformations individually on the triangle with (7)
coordinates (3,5) (6,2) and (4,7).
- (a) Translate by 5 units in x and -2 units in y direction.
- (b) Scale by 2 units in the x and y direction
- (c) Rotate by 45 degrees in clockwise.
- b) What are the steps required to fill a region using 4 connected boundary fill method. (7)
Explain the situations where 4-connected method fails to successfully fill the region.

OR

- 14 a) Explain the significance of homogeneous coordinate system. A polygon is defined (7)
by the pixel coordinates (0,0), (10,0), (15,5) and (5,5). It is subject to the following transformation. Scaling along the x and y axes with scaling factors of 1.5 and 0.5 respectively and a subsequent translation along the x and y -axes with 5 and 10 units respectively. Obtain the composite transformation matrix in homogeneous coordinates.
- b) Explain in detail about three-dimensional scaling and rotation transformation. (7)

Module III

- 15 a) Explain and illustrate the Sutherland- Hodgeman polygon clipping algorithm. (7)
- b) Given a clipping window A (-20, -20), B (40, -20), C (40, 30) and D (-20, 30). (7)
Using Cohen Sutherland line clipping algorithm, find the visible portion of the line segment joining the points P (-30, 20) and Q (60, -10).

OR

- 16 a) Explain in detail the scanline algorithm for visible surface detection by pointing (7)
out the data structures used in the algorithm.
- b) What is meant by projection? Define between parallel and perspective projection. (7)

Module IV

- 17 a) With a neat diagram, Explain the fundamental steps in image processing. (7)

- b) Define spatial resolution and gray level resolution of an image. How the image gets affected when (a) spatial resolution is reduced (ii) insufficient number of gray levels are used. (7)

OR

- 18 a) Explain the basic components of an image processing system with the help of a neat diagram. (7)
- b) Consider the image segment and compute the length of the shortest 4, 8 and m-path between p and q by considering two sets of values for V. (7)
- (i) $V = \{0, 1, 2\}$
- (ii) $V = \{1, 2\}$

If a particular path does not exist explain the reason for the above two cases of V

	3	4	1	2	0
	0	1	0	4	2(q)
	2	2	3	1	4
	3	0	4	2	1
(p)	1	2	0	3	4

Module V

- 19 a) A 5x5, 3 bits/pixel original image is given by (7)

4	4	4	4	4
3	4	5	4	3
3	5	5	5	3
3	4	5	4	3
4	4	4	4	4

- (i) Apply histogram equalization to the image by rounding the resulting image pixels to integers.
- (ii) Sketch the histogram of the original image and the histogram-equalized image.

- b) Explain the following grey level transformation functions (i) Power law transformation (ii) Log transformation (7)

OR

- 20 a) What is meant by edge detection in digital images? Describe Robert's and Sobel edge detectors. (7)
- b) Explain region splitting and merging algorithm for image segmentation. (7)
